

Pregunta 1

$$f(x) = \frac{\ln(1 + 2x^2 + x^4)^x (1 - \ln(x^2 + 1))}{(x^2 + 1)^3} \arctan\left(\frac{\ln(1 + x^2)}{1 + x^2}\right)$$

$$I = \int \frac{\ln((1 + x^2)^2)^x (1 - \ln(1 + x^2))}{(x^2 + 1)^3} \arctan\left(\frac{\ln(1 + x^2)}{1 + x^2}\right) dx$$

$$I = \int \frac{2x \ln(1 + x^2) (1 - \ln(1 + x^2))}{(1 + x^2)^3} \arctan\left(\frac{\ln(1 + x^2)}{1 + x^2}\right) dx$$

$$\text{Sea } u = \frac{\ln(1 + x^2)}{1 + x^2} \implies du = \frac{\frac{2x}{1+x^2}(1 + x^2) - 2x \ln(1 + x^2)}{(1 + x^2)^2} dx$$

$$du = \frac{2x(1 - \ln(1 + x^2))}{(1 + x^2)^2} dx$$

Reemplazando en la integral original:

$$I = \int u \arctan(u) du$$

$$\text{Integración por partes: } w = \arctan(u) \implies dw = \frac{1}{1 + u^2} du$$

$$dv = u du \implies v = \frac{u^2}{2}$$

$$I = \frac{u^2}{2} \arctan(u) - \frac{1}{2} \int \frac{u^2}{1 + u^2} du$$

$$I = \frac{u^2}{2} \arctan(u) - \frac{1}{2} \int \left(1 - \frac{1}{1 + u^2}\right) du$$

$$I = \frac{u^2}{2} \arctan(u) - \frac{1}{2}(u - \arctan(u)) + C$$

$$I = \frac{1}{2}(u^2 + 1) \arctan(u) - \frac{1}{2}u + C$$

Retornando a la variable x :

$$I = \frac{1}{2} \left(\left(\frac{\ln(1 + x^2)}{1 + x^2} \right)^2 + 1 \right) \arctan\left(\frac{\ln(1 + x^2)}{1 + x^2}\right) - \frac{\ln(1 + x^2)}{2(1 + x^2)} + C$$

Pregunta 2

$$I = \int \frac{1 - x \cos(a)}{1 - 2x \cos(a) + x^2} dx$$

Completando cuadrados y separando el numerador:

$$I = \int \frac{1 - x \cos(a)}{(x - \cos(a))^2 + \sin^2(a)} dx$$

$$\text{Sea } u = x - \cos(a) \implies x = u + \cos(a) \implies dx = du$$

$$I = \int \frac{1 - (u + \cos(a)) \cos(a)}{u^2 + \sin^2(a)} du$$

$$I = \int \frac{1 - u \cos(a) - \cos^2(a)}{u^2 + \sin^2(a)} du = \int \frac{\sin^2(a) - u \cos(a)}{u^2 + \sin^2(a)} du$$

$$I = \sin^2(a) \int \frac{1}{u^2 + \sin^2(a)} du - \cos(a) \int \frac{u}{u^2 + \sin^2(a)} du$$

$$I = \sin^2(a) \left(\frac{1}{\sin(a)} \arctan \left(\frac{u}{\sin(a)} \right) \right) - \frac{\cos(a)}{2} \ln(u^2 + \sin^2(a)) + C$$

$$I = \sin(a) \arctan \left(\frac{u}{\sin(a)} \right) - \frac{\cos(a)}{2} \ln(u^2 + \sin^2(a)) + C$$

Retornando a la variable x :

$$I = \sin(a) \arctan \left(\frac{x - \cos(a)}{\sin(a)} \right) - \frac{\cos(a)}{2} \ln(1 - 2x \cos(a) + x^2) + C$$

Pregunta 3

$$I = \int \sqrt{e^{3x}} \arctan \left(2\sqrt{\frac{e^x}{1-e^x}} \right) dx$$

Por integración por partes: $w = \arctan \left(2\sqrt{\frac{e^x}{1-e^x}} \right)$

$$dw = \frac{1}{1 + 4\frac{e^x}{1-e^x}} \cdot 2 \cdot \frac{1}{2\sqrt{\frac{e^x}{1-e^x}}} \cdot \frac{e^x(1-e^x) - e^x(-e^x)}{(1-e^x)^2} dx$$

$$dw = \frac{1-e^x}{1+3e^x} \cdot \sqrt{\frac{1-e^x}{e^x}} \cdot \frac{e^x}{(1-e^x)^2} dx = \frac{e^{x/2}}{(1+3e^x)\sqrt{1-e^x}} dx$$

$$dv = e^{3x/2} dx \implies v = \frac{2}{3}e^{3x/2}$$

$$I = \frac{2}{3}e^{3x/2} \arctan \left(2\sqrt{\frac{e^x}{1-e^x}} \right) - \frac{2}{3} \int \frac{e^{2x}}{(1+3e^x)\sqrt{1-e^x}} dx$$

Para la integral resultante, sea $t = \sqrt{1-e^x} \implies e^x = 1-t^2 \implies dx = \frac{-2t}{1-t^2} dt$

$$I_1 = \int \frac{(1-t^2)^2}{(1+3(1-t^2))t} \left(\frac{-2t}{1-t^2} \right) dt = -2 \int \frac{1-t^2}{4-3t^2} dt$$

$$I_1 = -2 \int \left(\frac{1}{3} - \frac{1}{3(4-3t^2)} \right) dt = -\frac{2}{3}t + \frac{2}{3} \int \frac{1}{4-3t^2} dt$$

$$I_1 = -\frac{2}{3}t + \frac{2}{3} \left(\frac{1}{4\sqrt{3}} \ln \left| \frac{2+\sqrt{3}t}{2-\sqrt{3}t} \right| \right) = -\frac{2}{3}t + \frac{1}{6\sqrt{3}} \ln \left| \frac{2+\sqrt{3}t}{2-\sqrt{3}t} \right|$$

Reemplazando en I :

$$I = \frac{2}{3}e^{3x/2} \arctan \left(2\sqrt{\frac{e^x}{1-e^x}} \right) + \frac{4}{9}\sqrt{1-e^x} - \frac{\sqrt{3}}{27} \ln \left| \frac{2+\sqrt{3(1-e^x)}}{2-\sqrt{3(1-e^x)}} \right| + C$$

Pregunta 4

$$I = \int \frac{\sqrt{\cot(x^2 + 2x)}}{\sin(x^2 + 2x) \cos(x^2 + 2x)} (x + 1) dx$$

$$\text{Sea } u = x^2 + 2x \implies du = 2(x + 1) dx \implies (x + 1) dx = \frac{du}{2}$$

$$I = \frac{1}{2} \int \frac{\sqrt{\cot(u)}}{\sin(u) \cos(u)} du$$

$$\text{Usando la identidad } \frac{1}{\sin(u) \cos(u)} = \frac{\csc^2(u)}{\cot(u)} :$$

$$I = \frac{1}{2} \int \frac{\sqrt{\cot(u)} \csc^2(u)}{\cot(u)} du = \frac{1}{2} \int \cot^{-1/2}(u) \csc^2(u) du$$

$$\text{Sea } v = \cot(u) \implies dv = -\csc^2(u) du$$

$$I = -\frac{1}{2} \int v^{-1/2} dv = -\frac{1}{2} (2v^{1/2}) + C = -v^{1/2} + C$$

Retornando a la variable x :

$$\boxed{I = -\sqrt{\cot(x^2 + 2x)} + C}$$

Pregunta 5

$$I = \int \frac{\sin^4(4x)}{\tan(x) + \tan(3x)} dx$$

Simplificando el denominador:

$$\tan(x) + \tan(3x) = \frac{\sin(x)}{\cos(x)} + \frac{\sin(3x)}{\cos(3x)} = \frac{\sin(x)\cos(3x) + \cos(x)\sin(3x)}{\cos(x)\cos(3x)} = \frac{\sin(4x)}{\cos(x)\cos(3x)}$$

$$I = \int \frac{\sin^4(4x)\cos(x)\cos(3x)}{\sin(4x)} dx = \int \sin^3(4x)\cos(x)\cos(3x) dx$$

Transformando producto a suma: $\cos(x)\cos(3x) = \frac{1}{2}(\cos(4x) + \cos(2x))$

$$I = \frac{1}{2} \int \sin^3(4x)\cos(4x) dx + \frac{1}{2} \int \sin^3(4x)\cos(2x) dx$$

Para la primera integral, sea $u = \sin(4x) \implies du = 4\cos(4x) dx$:

$$I_1 = \frac{1}{8} \int u^3 du = \frac{u^4}{32} = \frac{\sin^4(4x)}{32}$$

Para la segunda integral, usamos $\sin(4x) = 2\sin(2x)\cos(2x)$:

$$I_2 = \frac{1}{2} \int 8\sin^3(2x)\cos^3(2x)\cos(2x) dx = 4 \int \sin^3(2x)\cos^4(2x) dx$$

Sea $v = \cos(2x) \implies dv = -2\sin(2x) dx$ y $\sin^2(2x) = 1 - v^2$:

$$I_2 = -2 \int (1 - v^2)v^4 dv = -2 \int (v^4 - v^6) dv = -2 \left(\frac{v^5}{5} - \frac{v^7}{7} \right)$$

$$I_2 = -\frac{2}{5} \cos^5(2x) + \frac{2}{7} \cos^7(2x)$$

$$I = \frac{\sin^4(4x)}{32} - \frac{2}{5} \cos^5(2x) + \frac{2}{7} \cos^7(2x) + C$$

Pregunta 6

$$I = \int \frac{e^{24x}}{(e^{8x} + 2)^3} dx$$

$$\text{Sea } u = e^{8x} + 2 \implies e^{8x} = u - 2 \implies du = 8e^{8x} dx$$

$$\text{Reescribimos el numerador: } e^{24x} dx = e^{16x} e^{8x} dx = (u - 2)^2 \frac{du}{8}$$

$$I = \frac{1}{8} \int \frac{(u - 2)^2}{u^3} du = \frac{1}{8} \int \frac{u^2 - 4u + 4}{u^3} du$$

$$I = \frac{1}{8} \int (u^{-1} - 4u^{-2} + 4u^{-3}) du$$

$$I = \frac{1}{8} \left(\ln |u| + \frac{4}{u} - \frac{2}{u^2} \right) + C$$

Retornando a la variable x :

$$I = \frac{1}{8} \ln(e^{8x} + 2) + \frac{1}{2(e^{8x} + 2)} - \frac{1}{4(e^{8x} + 2)^2} + C$$

Pregunta 7

$$I = \int \frac{(\tan^3(x) + 3\tan(x) - 1)\sec^2(x)}{(\tan^2(x) - 4\tan(x) + 13)^3} dx$$

$$\text{Sea } u = \tan(x) \implies du = \sec^2(x) dx$$

$$I = \int \frac{u^3 + 3u - 1}{(u^2 - 4u + 13)^3} du$$

$$\text{Completando cuadrados en el denominador: } u^2 - 4u + 13 = (u - 2)^2 + 9$$

$$\text{Sea } t = u - 2 \implies u = t + 2 \implies du = dt$$

$$\text{El numerador es: } (t + 2)^3 + 3(t + 2) - 1 = t^3 + 6t^2 + 15t + 13 = t(t^2 + 9) + 6t^2 + 6t + 13$$

$$I = \int \frac{t(t^2 + 9) + 6t^2 + 6t + 13}{(t^2 + 9)^3} dt = \int \frac{t}{(t^2 + 9)^2} dt + \int \frac{6t}{(t^2 + 9)^3} dt + \int \frac{6t^2 + 13}{(t^2 + 9)^3} dt$$

$$I_1 = \int \frac{t}{(t^2 + 9)^2} dt = -\frac{1}{2(t^2 + 9)}$$

$$I_2 = \int \frac{6t}{(t^2 + 9)^3} dt = -\frac{3}{2(t^2 + 9)^2}$$

$$\text{Para } I_3, \text{ sea } t = 3 \tan \theta \implies dt = 3 \sec^2 \theta d\theta \quad \text{y} \quad t^2 + 9 = 9 \sec^2 \theta :$$

$$I_3 = \int \frac{6(9 \tan^2 \theta) + 13}{(9 \sec^2 \theta)^3} (3 \sec^2 \theta) d\theta = \frac{1}{243} \int \frac{54 \tan^2 \theta + 13}{\sec^4 \theta} d\theta$$

$$I_3 = \frac{1}{243} \int (54 \sin^2 \theta \cos^2 \theta + 13 \cos^4 \theta) d\theta$$

$$I_3 = \frac{1}{243} \int \left(\frac{27}{2} (1 - \cos(4\theta)) + \frac{13}{8} (3 + 4 \cos(2\theta) + \cos(4\theta)) \right) d\theta$$

$$I_3 = \frac{1}{1944} \int (93 + 52 \cos(2\theta) - 41 \cos(4\theta)) d\theta$$

$$I_3 = \frac{31}{648} \theta + \frac{13}{972} \sin(2\theta) - \frac{41}{7776} \sin(4\theta)$$

$$\text{Expresando en términos de } t \text{ usando } \theta = \arctan(t/3) :$$

$$\sin(2\theta) = \frac{6t}{t^2 + 9} \quad ; \quad \sin(4\theta) = \frac{12t(9 - t^2)}{(t^2 + 9)^2}$$

$$I_3 = \frac{31}{648} \arctan\left(\frac{t}{3}\right) + \frac{13t}{162(t^2 + 9)} - \frac{41t(9 - t^2)}{648(t^2 + 9)^2}$$

$$I = -\frac{1}{2(t^2 + 9)} - \frac{3}{2(t^2 + 9)^2} + \frac{31}{648} \arctan\left(\frac{t}{3}\right) + \frac{13t}{162(t^2 + 9)} - \frac{41t(9 - t^2)}{648(t^2 + 9)^2} + C$$

Pregunta 8

8. Determine la function $y = f(x)$, de tal manera que la pendiente de la recta tangente a la grafica de $f(x)$ en el punto $(x, f(x))$ es la siguiente expresion :

$$f'(x) = \frac{6 \sin^3 x \cos x - \cos^3 x + 0,5 \sin 2x + 5 \cos x}{\sin^5 x - 2 \sin^4 x + \sin^3 x + \cos^2 x + 2 \sin x - 2}$$

Integrando la función: $f(x) = \int f'(x) dx$

Usando $\cos^2 x = 1 - \sin^2 x$ y $0,5 \sin(2x) = \sin x \cos x$:

$$f(x) = \int \frac{6 \sin^3 x \cos x - \cos x(1 - \sin^2 x) + \sin x \cos x + 5 \cos x}{\sin^5 x - 2 \sin^4 x + \sin^3 x + 1 - \sin^2 x + 2 \sin x - 2} dx$$

$$f(x) = \int \frac{\cos x(6 \sin^3 x + \sin^2 x + \sin x + 4)}{\sin^5 x - 2 \sin^4 x + \sin^3 x - \sin^2 x + 2 \sin x - 1} dx$$

Sea $u = \sin x \implies du = \cos x dx$

$$f(x) = \int \frac{6u^3 + u^2 + u + 4}{u^5 - 2u^4 + u^3 - u^2 + 2u - 1} du$$

Factorizando el denominador: $u^5 - 2u^4 + u^3 - u^2 + 2u - 1 = (u - 1)^3(u^2 + u + 1)$

Descomposición en fracciones parciales:

$$\frac{6u^3 + u^2 + u + 4}{(u - 1)^3(u^2 + u + 1)} = \frac{2}{u - 1} + \frac{3}{(u - 1)^2} + \frac{4}{(u - 1)^3} - \frac{2u + 1}{u^2 + u + 1}$$

$$f(x) = \int \left(\frac{2}{u - 1} + \frac{3}{(u - 1)^2} + \frac{4}{(u - 1)^3} - \frac{2u + 1}{u^2 + u + 1} \right) du$$

$$f(x) = 2 \ln |u - 1| - \frac{3}{u - 1} - \frac{2}{(u - 1)^2} - \ln(u^2 + u + 1) + C$$

$$f(x) = \ln \left(\frac{(u - 1)^2}{u^2 + u + 1} \right) + \frac{3}{1 - u} - \frac{2}{(1 - u)^2} + C$$

Retornando a x :

$$f(x) = \ln \left(\frac{(1 - \sin x)^2}{\sin^2 x + \sin x + 1} \right) + \frac{3}{1 - \sin x} - \frac{2}{(1 - \sin x)^2} + C$$